1.25 Conics: Transformations

Worked Solutions

1 a)

$y^2 = 4(\frac{1}{3}x)$

$$\therefore y^2 = \frac{4}{3}x$$

1 b)

$(4y)^2 = 4x$

$$\therefore y^2 = \frac{1}{4}x$$

1 c)

$(y+1)^2 = 4(x-2)$

1 d)

$(-y)^2 = 4x$

$$\therefore y^2 = 4x$$

1 e)

$x^2 = 4y$

2 a)

$(2x)^2 + 3y^2 = 3$

$$\therefore 4x^2 + 3y^2 = 3$$

2 b)

$x^2 + 3(\frac{1}{5}y)^2 = 3$

$$\therefore x^2 + \frac{3}{25}y^2 = 3$$

2 c)

$(x+3)^2 + 3(y-4)^2 = 3$

2 d)

$x^2 + 3(-y)^2 = 3$

$$\therefore x^2 + 3y^2 = 3$$

2 e)

$y^2 + 3x^2 = 3$

3 a)

$(\frac{1}{3}x)y = 12 \quad \therefore xy = 36$

3 b)

$x(\frac{1}{2}y) = 12 \quad \therefore xy = 24$

3 c)

$xy = 12 \quad \therefore (x-2)(y-2) = 12$

3 d)

$x(-y) = 12 \quad \therefore xy = -12$

3 e)

$(-y)(-x) = 12 \quad \therefore xy = 12$

4 a)

$\frac{(4x)^2}{4} - \frac{y^2}{5} = 1$

$$\therefore 4x^2 - \frac{y^2}{5} = 1$$

4 b)

$\frac{x^2}{4} - \frac{(\frac{1}{2}y)^2}{5} = 1$

$$\therefore \frac{x^2}{4} - \frac{y^2}{20} = 1$$

4 c)

$\frac{(x-3)^2}{4} - \frac{(y+5)^2}{5} = 1$

4 d)

$\frac{x^2}{4} - \frac{y^2}{5} = 1$

4 e)

$\frac{y^2}{4} - \frac{x^2}{5} = 1$

5 a)

$\frac{(\frac{1}{2}x)^2}{3} + \frac{y^2}{16} = 1$

$$\therefore \frac{x^2}{12} + \frac{y^2}{16} = 1, \text{ Ellipse}$$

5 b)

$\frac{(\frac{1}{2}x)^2}{2} - \frac{y^2}{10} = 1$

$$\therefore \frac{x^2}{8} - \frac{y^2}{10} = 1, \text{ Hyperbola}$$

5 c)

$(\frac{1}{2}x)y = 14$

$$\therefore xy = 28, \text{ Rectangular Hyperbola}$$

5 d)

$(\frac{1}{2}x)^2 - y^2 = 4$

$$\therefore \frac{x^2}{4} - y^2 = 4, \text{ Hyperbola}$$

6 a)

$\frac{x^2}{18} + \frac{(\frac{1}{3}y)^2}{2} = 1$

$$\therefore x^2 + y^2 = 18, \text{ Circle}$$

6 b)

$\frac{x^2}{45} - \frac{(\frac{1}{3}y)^2}{5} = 1$

$$\therefore x^2 - y^2 = 45, \text{ Rectangular Hyperbola}$$

6 c)

$(\frac{1}{3}y)^2 = -3x$

$$\therefore y^2 = -27x, \text{ Parabola}$$

6 d)

$x^2 - (\frac{1}{3}y)^2 = 2$

$$\therefore \frac{x^2}{2} - \frac{y^2}{18} = 1, \text{ Hyperbola}$$

7 a)

$(\frac{1}{2}x)^2 + y^2 = 4$

$$\therefore \frac{x^2}{16} + \frac{y^2}{4} = 1$$

7 b)

$x^2 + (\frac{1}{2}y)^2 = 4$

$$\therefore \frac{x^2}{4} + \frac{y^2}{16} = 1$$

8 a)

$x^2 = \frac{1}{2}y$

$$\therefore y = 2x^2$$

8 b)

$7x^2 + 3y^2 = 21$

8 c)

$xy = -6$

8 d)

$\frac{y^2}{8} - \frac{x^2}{4} = 1$

9 a)

$(-x)^2 = \frac{1}{4}(-y)$

$$\therefore x^2 = -\frac{1}{4}y$$

9 b)

$2(-y)^2 + 5(-x)^2 = 10$

$$\therefore 2y^2 + 5x^2 = 10$$

9 c)

$(-y)(-x) = \frac{1}{5}$

$$\therefore xy = \frac{1}{5}$$

9 d)

$\frac{(-y)^2}{3} - \frac{(-x)^2}{25} = 1$

$$\therefore \frac{y^2}{3} - \frac{x^2}{25} = 1$$

10 a)

Reflection in $y = x$

$$\frac{x^2}{8} + \frac{y^2}{4} = 1$$

10 b)

Reflection in $y = -x$

$$\frac{(-y)^2}{4} + \frac{(-x)^2}{8} = 1$$

$$\therefore \frac{x^2}{8} + \frac{y^2}{4} = 1$$

11 a)

Reflection in $y = x$

$$\frac{y^2}{12} - \frac{x^2}{4} = 1$$

11 b)

$x = \pm\frac{2}{\sqrt{12}}y$

So $y = \sqrt{3}x$ and $y = -\sqrt{3}x$

11 c)

A stretch, parallel to the y-axis, scale factor 2

$$\frac{x^2}{12} - \frac{(\frac{1}{2}y)^2}{4} = 1 \quad \therefore \frac{x^2}{12} - \frac{y^2}{16} = 1$$

11 d)

$y = \pm\frac{4}{\sqrt{12}}x$

So $y = \frac{2\sqrt{3}}{3}x$ and $y = -\frac{2\sqrt{3}}{3}x$

12 a)

Translation $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$

12 b)

Translation $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$

12 c)

Translation $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$

12 d)

A stretch, parallel to the x-axis, scale factor $\frac{1}{2}$

12 e)

A stretch, parallel to the x-axis, scale factor 3

12 f)

$\frac{x^2}{16} - (y+2)^2 = 16$

Translation $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$

12 g)

$\frac{(x+5)^2}{8} - \frac{(y-1)^2}{2} = 1$

Translation $\begin{pmatrix} -5 \\ 1 \end{pmatrix}$

12 h)

$\frac{x^2}{16} - 18y^2 = 1$

A stretch, parallel to the y-axis, scale factor $\frac{1}{6}$

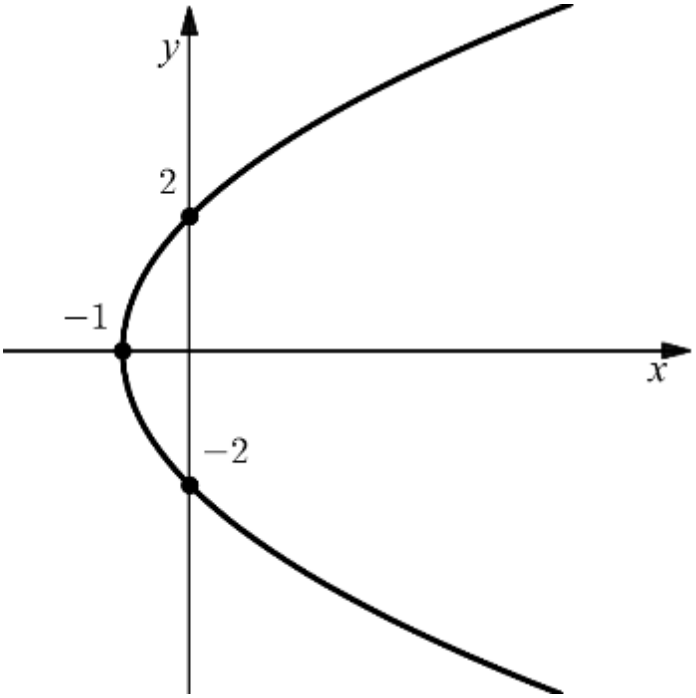
12 i)

$$\begin{aligned} 2x^2 - 12x - 3y^2 - 18y &= 21 \\ 2(x^2 - 6x) - 3(y^2 + 6y) &= 21 \\ 2((x-3)^2 - 9) - 3((y+3)^2 - 9) &= 21 \\ 2(x-3)^2 - 18 - 3(y+3)^2 + 27 &= 21 \\ 2(x-3)^2 - 3(y+3)^2 &= 12 \\ \frac{(x-3)^2}{6} - \frac{(y+3)^2}{4} &= 1 \end{aligned}$$

Translation $\begin{pmatrix} 3 \\ -3 \end{pmatrix}$

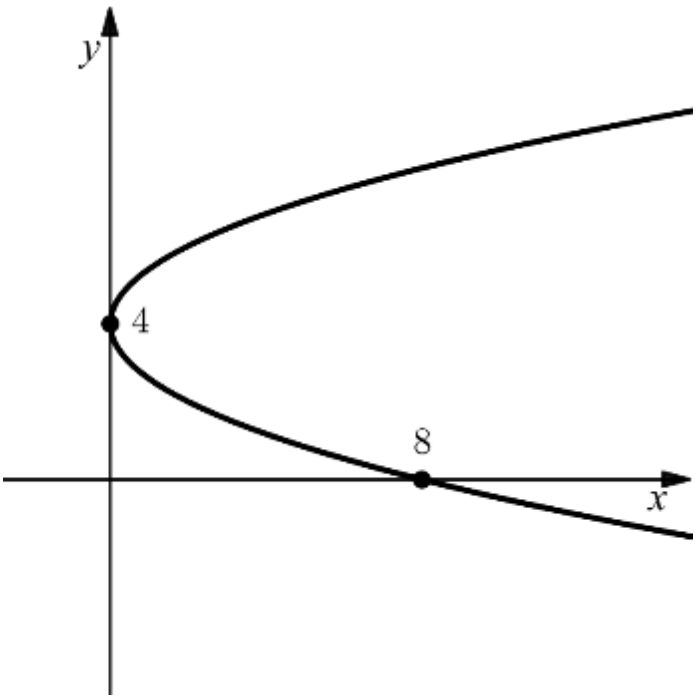
13 a)

When $x = 0$, $y = \pm 2$, when $y = 0$, $x = -1$



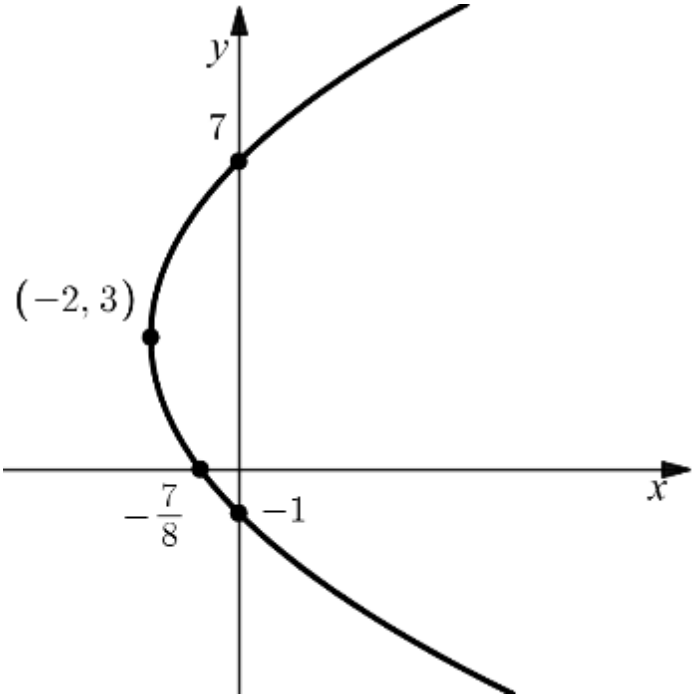
13 b)

When $x = 0$, $y = 4$, when $y = 0$, $x = -1$



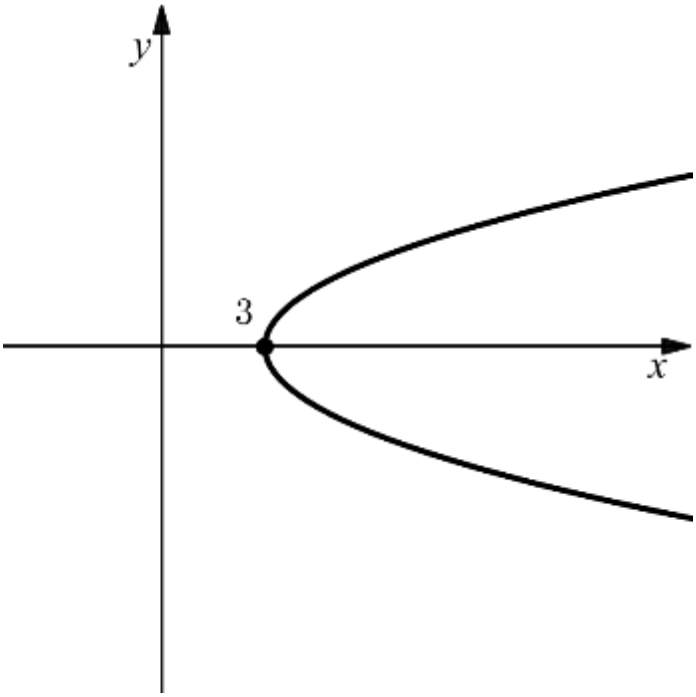
13 c)

When $x = 0$, $y = -1$ or $y = 7$, when $y = 0$, $x = -\frac{7}{8}$



13 d)

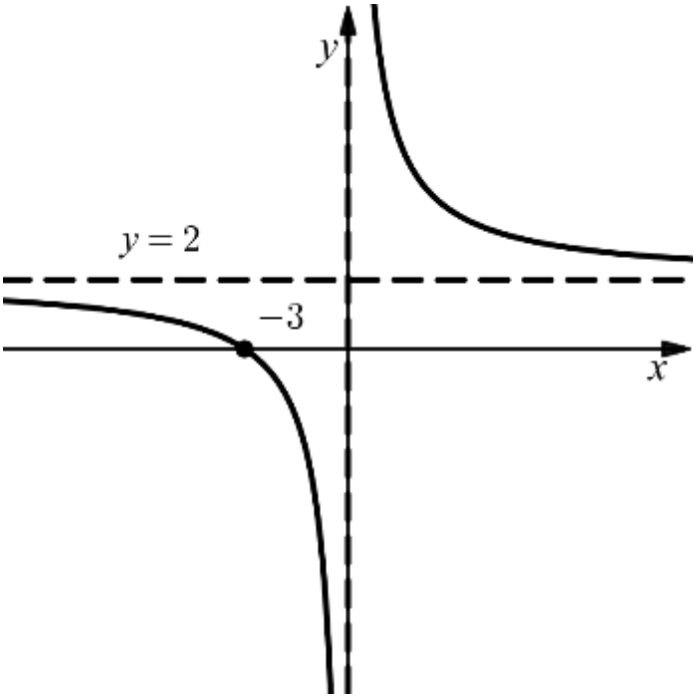
When $x = 0$, no solutions when $y = 0$, $x = 3$



14 a)

When $x = 0$, no solutions, when $y = 0$, $x = -3$

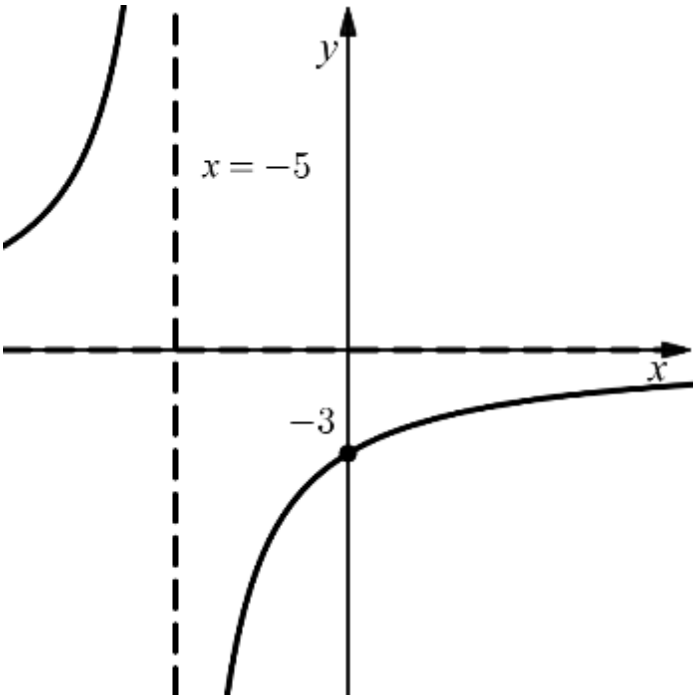
Asymptotes: $x = 0$, $y = 2$



14 b)

When $x = 0$, $y = -3$, when $y = 0$, no solutions

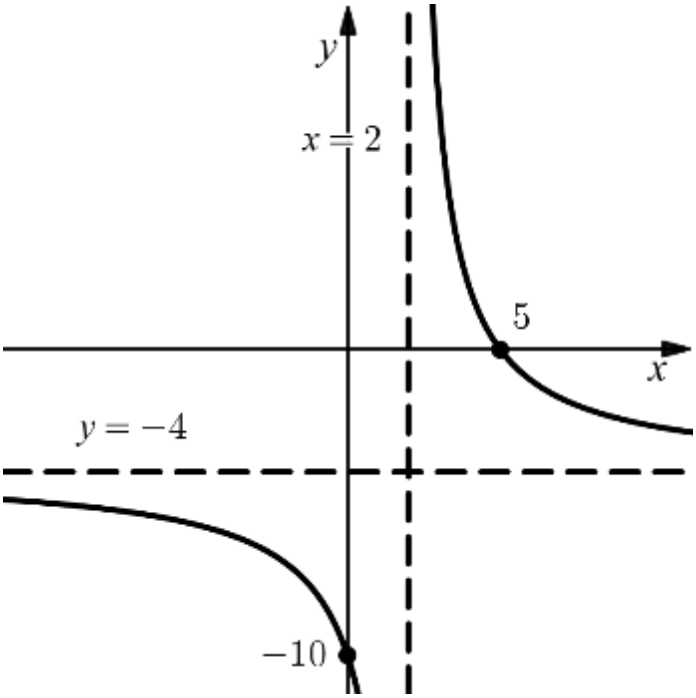
Asymptotes: $x = -5$, $y = 0$



14 c)

When $x = 0$, $y = -10$, When $y = 0$, $x = 5$

Asymptotes: $x = 2$, $y = -4$

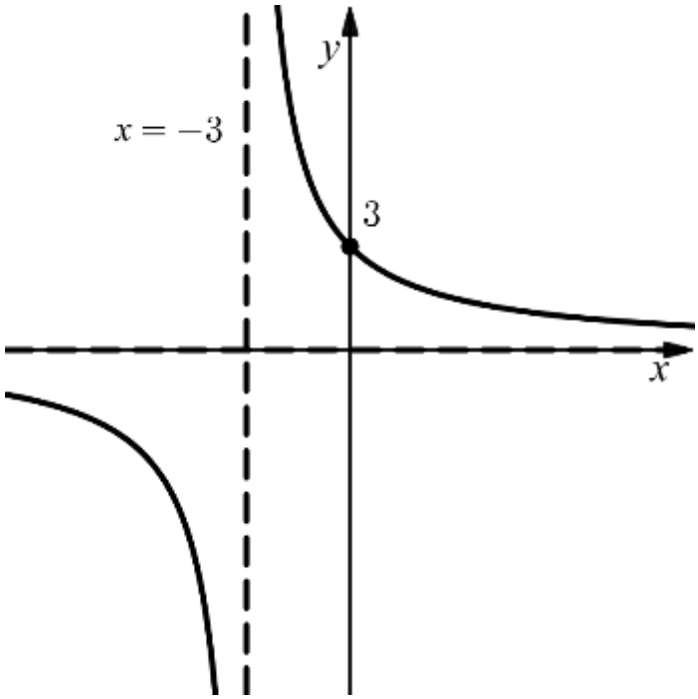


14 d)

$y(3 + x) = 9$

When $x = 0$, $y = 3$, when $y = 0$, no solutions

Asymptotes: $x = -3$, $y = 0$

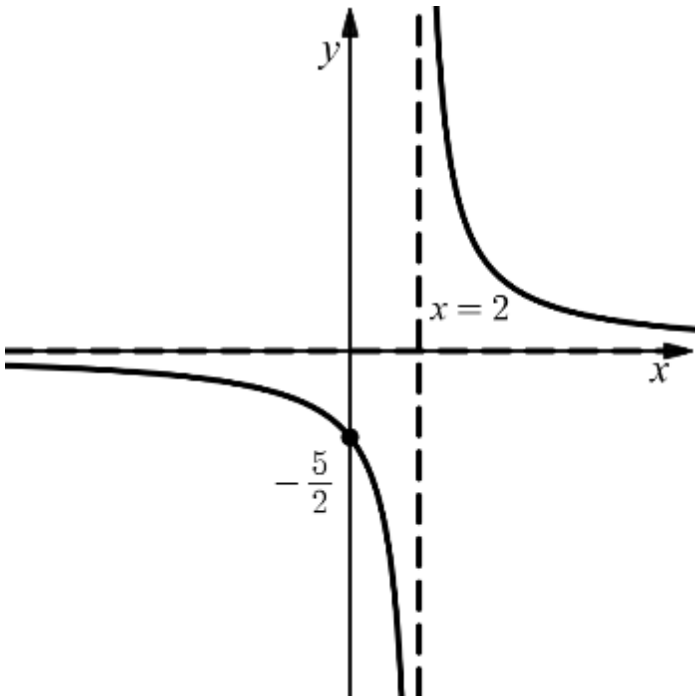


14 e)

$y(x-2)=5$

When $x=0$, $y=-\frac{5}{2}$, when $y=0$, no solutions

Asymptotes: $x=-3$, $y=0$



15 a)

$\frac{x^2}{25}+\frac{(y+1)^2}{5}=1$

Translation $\begin{pmatrix}0\\-1\end{pmatrix}$

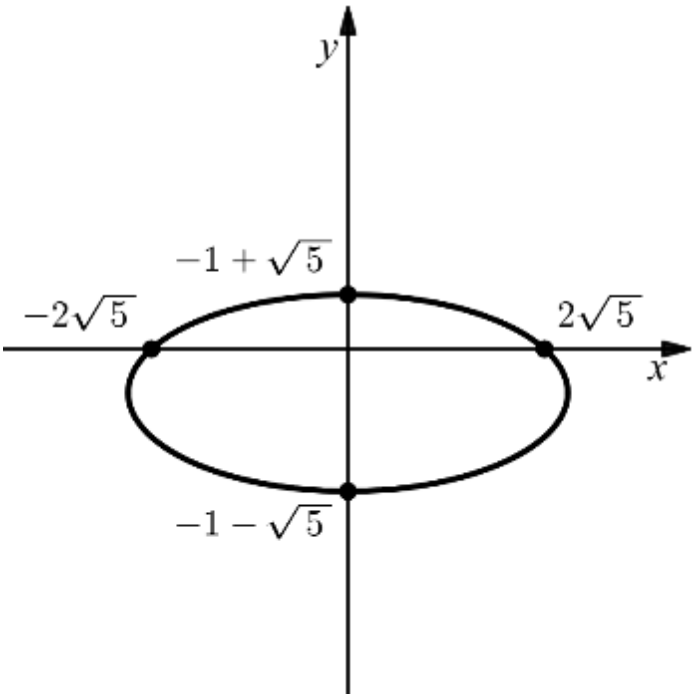
15 b)

When $x = 0$, $(y+1)^2 = 5$ so $y = -1 \pm \sqrt{5}$,

When $y = 0$, $x^2 = 20$ so $x = \pm 2\sqrt{5}$

$(0, -1-\sqrt{5})$, $(0, -1+\sqrt{5})$, $(2\sqrt{5}, 0)$ and $(-2\sqrt{5}, 0)$

15 c)



16 a)

Hyperbola

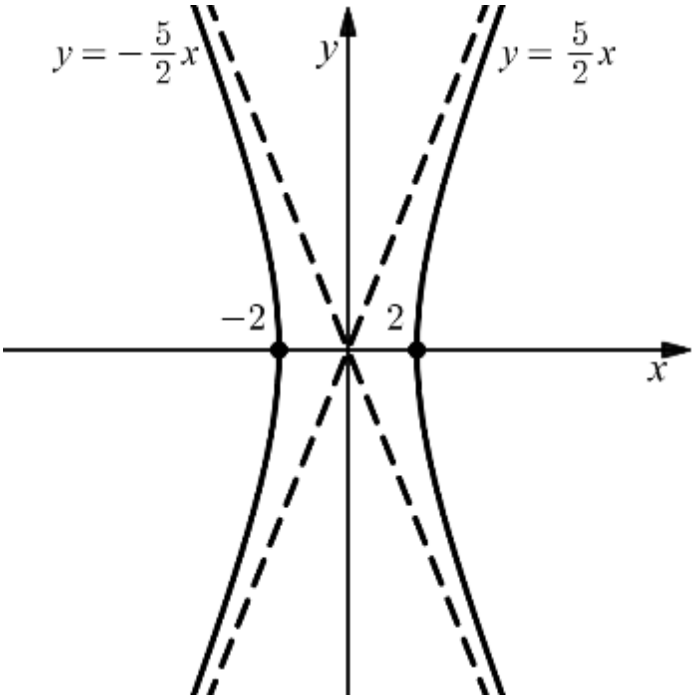
16 b)

$y = \frac{2\sqrt{3}}{3}x$ and $y = -\frac{2\sqrt{3}}{3}x$

16 c)

$y = \frac{2\sqrt{3}}{3}(x-4)$ and $y = -\frac{2\sqrt{3}}{3}(x-4)$

17 a)



17 b)

$k = 2$

17 c)

$y = \frac{5}{2}x$ and $y = -\frac{5}{2}x$ map to $y = \frac{5}{2}(x-2)$ and $y = -\frac{5}{2}(x-2)$

$y = \frac{5}{2}x - 5$ and $y = -\frac{5}{2}x + 5$

18 a)

When $x = 0$, $y = \pm 1$, when $y = 0$, $x = \pm 3$

$(0, -1)$, $(0, 1)$, $(3, 0)$ and $(-3, 0)$

18 b)

$$\begin{aligned} x^2 - 4x + 9y^2 &= 5 \\ (x-2)^2 - 4 + 9y^2 &= 5 \\ (x-2)^2 + 9y^2 &= 9 \\ \frac{(x-2)^2}{9} + y^2 &= 1 \end{aligned}$$

Translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

18 c)

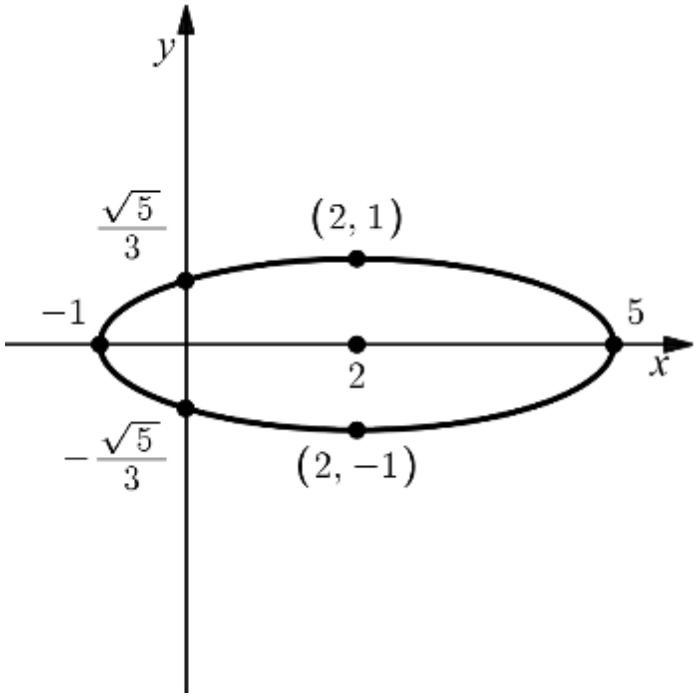
$(2, -1)$, $(2, 1)$, $(5, 0)$ and $(-1, 0)$

18 d)

When $x = 0$, $y = \pm\frac{\sqrt{5}}{3}$, when $y = 0$, $x = -1$, $x = 5$

$(0, -\frac{\sqrt{5}}{3})$, $(0, \frac{\sqrt{5}}{3})$, $(5, 0)$ and $(-1, 0)$

18 e)



19 a)

$y = \frac{2}{3}x$ and $y = -\frac{2}{3}x$

19 b)

Translation $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

19 c)

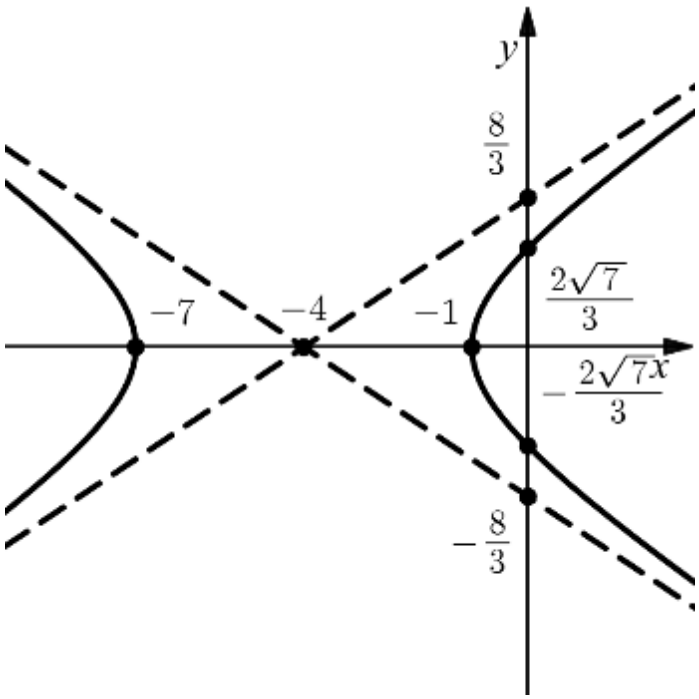
$y = \frac{2}{3}(x + 4)$ and $y = -\frac{2}{3}(x + 4)$

19 d)

When $x = 0$, $\frac{16}{9} - \frac{y^2}{4} = 1 \quad \therefore y = \pm\frac{2\sqrt{7}}{3}$, when $y = 0$, $(x + 4)^2 = 9 \quad \therefore x = -1$ or $x = -7$

$(0, \frac{2\sqrt{7}}{3})$, $(0, -\frac{2\sqrt{7}}{3})$, $(-1, 0)$ and $(-7, 0)$

19 e)



20 a)

$y - 2 = \frac{5}{2}x$ and $y - 2 = -\frac{5}{2}x$

So $y = \frac{5}{2}x + 2$ and $y = -\frac{5}{2}x + 2$

20 b)

Translation $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$

20 c)

$y + 5 = \frac{5}{2}x + 2$ and $y + 5 = -\frac{5}{2}x + 2$

So $y = \frac{5}{2}x - 3$ and $y = -\frac{5}{2}x - 3$

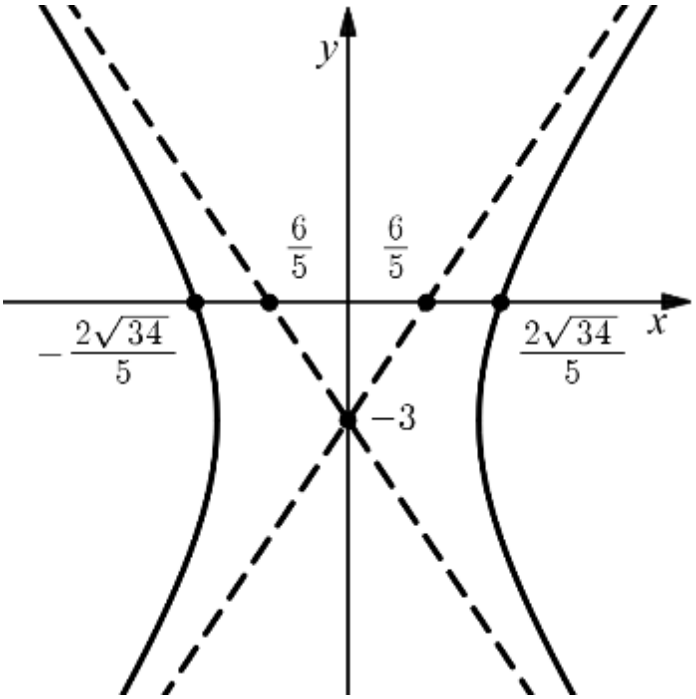
20 d)

When $x = 0$, $(y + 3)^2 = -25$ so no solutions

When $y = 0$, $\frac{x^2}{4} - \frac{9}{25} = 1 \quad \therefore x = \pm\frac{2\sqrt{34}}{5}$

$(\frac{2\sqrt{34}}{5}, 0)$ and $(-\frac{2\sqrt{34}}{5}, 0)$

20 e)



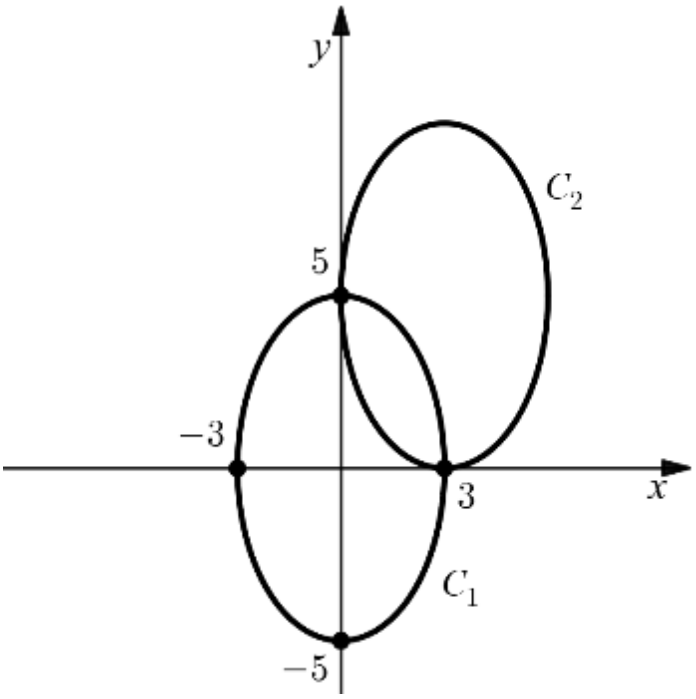
21 a)

$\begin{pmatrix} 3 \\ 5 \end{pmatrix}$

21 b)

$(3, 0)$ and $(0, 5)$

21 c)



21 d)

The tangent lines have gradient $\frac{5}{3}$

21 e)

For $y = \frac{5}{3}x + k$

With C_1

$$\begin{aligned}\frac{x^2}{9} + \frac{(\frac{5}{3}x + k)^2}{25} = 1 &\Rightarrow 25x^2 + 9(\tfrac{5}{3}x + k)^2 = 225 \\ &\Rightarrow 25x^2 + 9(\tfrac{25}{9}x^2 + \tfrac{10}{3}kx + k^2) = 225 \\ &\Rightarrow 25x^2 + 25x^2 + 30kx + 9k^2 = 225 \\ &\Rightarrow 50x^2 + 30kx + 9k^2 - 225 = 0\end{aligned}$$

Discriminant: $(30k)^2 - 4 \times 50(9k^2 - 225) = 0$

$$-900k^2 + 45000 = 0$$

$\therefore k = \pm\sqrt{50} = \pm 5\sqrt{2}$

So the tangent lines are $y = \frac{5}{3}x + 5\sqrt{2}$ and $y = \frac{5}{3}x - 5\sqrt{2}$

22 a)

When $x = 0$, $y = \pm\sqrt{3}$ and when $y = 0$ $x = \pm 2\sqrt{2}$

$(0, -\sqrt{3})$, $(0, \sqrt{3})$, $(2\sqrt{2}, 0)$ and $(-2\sqrt{2}, 0)$

22 b)

$\frac{(x-k)^2}{8} + \frac{y^2}{3} = 1$

22 c)

$$\begin{aligned}\frac{(x-k)^2}{8} + \frac{(2x+1)^2}{3} &= 1 \\ 3(x-k)^2 + 8(2x+1)^2 &= 24 \\ 3(x^2 - 2kx + k^2) + 8(4x^2 + 4x + 1) &= 24 \\ 3x^2 - 6kx + 3k^2 + 32x^2 + 32x + 8 &= 24 \\ 35x^2 + (32 - 6k)x + (3k^2 - 16) &= 0 \text{ as req.}\end{aligned}$$

22 d)

Tangent $\therefore b^2 - 4ac = 0$

$$\begin{aligned}(32 - 6k)^2 - 4 \times 35 \times (3k^2 - 16) &= 0 \\ 36k^2 - 384k + 1024 - 420k^2 + 2240 &= 0 \\ -384k^2 - 384k + 3264 &= 0 \\ 2k^2 + 2k - 17 &= 0 \\ k &= -\frac{1}{2} \pm \frac{\sqrt{35}}{2}\end{aligned}$$

So since k is positive $k = -\frac{1}{2} + \frac{\sqrt{35}}{2}$

23 a)

$(\frac{1}{3}x)^2 + y^2 = 10$

$$\therefore \frac{x^2}{9} + y^2 = 10$$

23 b)

$3(\frac{1}{3}x) - y = 10$

$$\therefore x - y = 10$$

23 c)

$$\frac{x^2}{9} + (x-10)^2 = 10$$
$$\frac{x^2}{9} + x^2 - 20x + 100 = 10$$
$$x^2 + 9x^2 - 180x + 900 = 90$$
$$x^2 - 18x + 81 = 0$$
$$b^2 - 4ac = (-18)^2 - 4 \times 1 \times 81 = 324 - 324 = 0$$

One solution, so the line is a tangent to the curve

They intersect when $x = 9$, $y = -1$

$$(9, -1)$$

23 d)

$\frac{x^2}{90} + \frac{y^2}{10} = 1$ and $x - y = 10$

